PROJECT PROPOSAL · MILESTONE 1

S&P 500 Stock Prediction Using Deep Learning and Sentiment Analysis

Project Title	S&P 500 Stock Prediction Using Deep Learning and Sentiment Analysis
Group Number	6
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Course	Applied Integrative Analytics Capstone Project (MBAI 5600G)

1. Executive Summary / Abstract

Predicting stock price movements in the US equity market is one of the most challenging and high-stakes problems in applied machine learning. The complexity arises from the non-linear relationships between price dynamics, trading volume, technical indicators, and the ever-present influence of market sentiment driven by financial news. This project addresses this challenge by developing a comprehensive, multi-model predictive analytics system targeting all S&P 500 constituent stocks, with two simultaneous outputs: the predicted next-day closing price (regression) and a binary buy or sell signal (classification) to support actionable investment decisions.

The proposed solution implements and rigorously compares five models to identify the best-performing approach for both price prediction and buy/sell signal generation. All five models are trained on the same feature set and evaluated under identical conditions to ensure a fair, reproducible comparison. The models are: (1) LSTM+GRU - a hybrid recurrent architecture combining long short-term memory and gated recurrent units, following the benchmark from our base research article; (2) Bidirectional LSTM (BiLSTM) - which processes sequences in both forward and backward temporal directions for richer context; (3) CNN+LSTM - a hybrid that uses convolutional layers for local pattern extraction followed by LSTM for temporal modelling; (4) Time Series Transformer - a self-attention-based model designed for multivariate time series with built-in interpretability; and (5) Random Forest - an interpretable ensemble baseline that captures non-linear feature interactions without deep learning. The model comparison is a central contribution of this project, with the best model identified based on both predictive accuracy metrics and real-world financial performance.

The feature engineering pipeline is a core contribution, incorporating technical indicators (RSI, MACD, Bollinger Bands, ATR, and others), lag features, rolling statistical features, and FinBERT-based sentiment-price interaction features. Financial performance is evaluated using profit/loss (P&L) simulation and the Sharpe Ratio to quantify real-world trading utility. Backtesting is conducted on the buy/sell signal predictions to validate strategy viability. All data is sourced from yahoo finance via the yfinance Python library. Regression performance is evaluated using MSE, MAE, RMSE, and MAPE, while classification is measured by directional accuracy, precision, recall, F1-score, and ROC-AUC.

2. Problem Statement & Project Objectives

2.1 Formal Problem Statement

Develop a dual-output predictive analytics system for S&P 500 stocks that simultaneously forecasts next-day closing price and generates a buy or sell trading signal, using advanced feature engineering (technical indicators, lag features, rolling statistics, and FinBERT-based sentiment-price interaction features derived from Yahoo Finance historical data and Yahoo Finance News), implemented through five models: LSTM+GRU, BiLSTM, CNN+LSTM, Time Series Transformer and Random Forest. The system targets a minimum directional accuracy of 60% for buy/sell signal classification and a minimum 10% improvement in RMSE over a naive baseline, validated through backtesting and financial metrics including P&L simulation and Sharpe Ratio.

2.2 SMART Analysis

Criterion	Attribute	Project Application
S	Specific	Predict S&P 500 next-day closing price (regression) and buy/sell signal (classification) using technical indicators, lag features, rolling stats, FinBERT sentiment-price interactions, and five distinct models across deep learning and ensemble approaches.
M	Measurable	Regression: approximately10%+ RMSE improvement over a naive baseline. Classification: 60%+ directional accuracy and F1-score ≥ 0.60. Financial: positive Sharpe Ratio > 1.0 and positive P&L in backtesting on the held-out test set.
A	Achievable	All data is freely accessible via yfinance. Tools (Python, Scikit-learn, PyTorch) are open-source. All five architectures are well-documented with strong results on financial time series in recent literature.
R	Relevant	Accurate price forecasting and reliable buy/sell signals with financial performance validation are directly applicable for retail investors, quantitative analysts, and algorithmic trading systems in the US equity market.
T	Time-Bound	Completed within the academic semester. Milestones: data collection & EDA (Weeks 1 - 2), feature engineering (Weeks 3 - 5), model training & tuning (Weeks 6 - 8), backtesting & evaluation (Weeks 9 - 10).

2.3 Key Objectives

- Collect, clean, and preprocess 10 years of historical OHLCV data for all S&P 500 constituent stocks using the yfinance Python API, including handling of missing trading days, stock splits, and dividend adjustments.
- Engineer a rich, expanded feature space comprising: technical indicators (RSI, MACD, Bollinger Bands, ATR, OBV, EMA, SMA, Stochastic Oscillator, ADX, CCI, Williams %R), lag features (1-day, 5-day, 10-day, 20-day), rolling statistical features (mean, std, min, max, skewness), and FinBERT-derived daily sentiment scores from Yahoo Finance News headlines.
- Apply enhanced feature engineering including calendar features (day of week, month, quarter), volume-price divergence indicators, and cross-sectional sector momentum to improve model signal quality.
- Create novel sentiment - price interaction features - such as sentiment score multiplied by price momentum or volume - to capture the combined effect of market mood and price dynamics.
- Train and evaluate all five models (LSTM+GRU, BiLSTM, CNN+LSTM, Time Series Transformer, Random Forest) for both regression and classification tasks, comparing their strengths across different market regimes.
- Conduct backtesting on the buy/sell signal predictions over the held-out test period, simulating a realistic trading strategy to measure real-world viability.
- Compute financial performance metrics including Profit & Loss (P&L) simulation and the Sharpe Ratio to quantify risk-adjusted returns beyond standard ML metrics.

- Compare all models using regression metrics (MSE, MAE, RMSE, MAPE) and classification metrics (directional accuracy, precision, recall, F1-score, ROC-AUC) on an 60/20/20 train - validate - test split with time-series-aware cross-validation.
- Analyse feature importance using SHAP values for Random Forest and attention weights from the Transformer to identify which features drive accurate predictions.
- Document all experimental results, model comparisons, and backtesting insights in a reproducible Python codebase with comprehensive Jupyter notebooks.

2.4 Base Research Article

Reference Article:
A. Farhadi, A. Zamanifar, A. Alipour, A. Taheri, and M. Asadolahi, “A hybrid LSTM-GRU model for stock price prediction,” IEEE Access, vol. 13, pp. 117594 - 117618, Jul. 2025, doi: https://doi.org/10.1109/access.2025.3586558.

Why This Article Was Selected
This 2025 IEEE Access article was selected because it directly benchmarks the LSTM+GRU hybrid architecture that this project implements as one of its five models. The paper rigorously evaluates prediction using the same technical indicator categories (trend, momentum, volatility, volume) adopted here, and validates MSE, MAE, RMSE, and MAPE as evaluation metrics. The authors demonstrate the importance of a comprehensive feature space over raw price data alone - a finding that directly motivates this project's advanced feature engineering approach.

How This Project Differs and Extends
This project extends the base paper by implementing the LSTM+GRU architecture as one of five complementary models (rather than the sole approach), targeting the US S&P 500 market instead of the Tehran Stock Exchange, adding FinBERT sentiment analysis, introducing financial metrics (P&L simulation, Sharpe Ratio), and validating strategies through backtesting. The table below summarizes the key differences:

Dimension	Farhadi et al. [1]	This Project
Market	Tehran Stock Exchange (8 stocks)	US S&P 500 (500 stocks)
Core Models	Hybrid LSTM-GRU only	LSTM+GRU, BiLSTM, CNN+LSTM, Time Series Transformer, Random Forest
Sentiment Analysis	None	FinBERT on Yahoo Finance News
Interaction Features	None	Sentiment × price momentum, sentiment × volume
Output	Price prediction only (regression)	Price prediction + Buy/Sell signal (dual output)
Financial Validation	Not addressed	P&L simulation, Sharpe Ratio, Backtesting
Interpretability	Not addressed	SHAP values + Transformer attention weights

3. Project Scope

This section defines exactly what is within and outside the boundaries of this project to ensure clarity of deliverables and feasibility within the available timeline.

3.1 Included Components

Models & Algorithms

- LSTM+GRU Hybrid - primary recurrent deep learning model combining long short-term memory and gated recurrent units, following the architecture from the base research article
- Bidirectional LSTM (BiLSTM) - processes sequences in both temporal directions for richer pattern capture
- CNN+LSTM Hybrid - uses convolutional layers for local feature extraction followed by LSTM for temporal modelling
- Time Series Transformer - self-attention-based model for multivariate time series with built-in interpretability through attention weights
- Random Forest Regressor and Classifier - interpretable ensemble baseline capturing non-linear feature interactions
- FinBERT - financial-domain BERT model for sentiment scoring of Yahoo Finance News headlines

Datasets

- Yahoo Finance historical OHLCV price data via yfinance - all S&P 500 constituents, 2015 to 2024 (~10 years of daily data)
- Yahoo Finance News headlines per ticker - used exclusively for FinBERT sentiment feature extraction

Enhanced Feature Engineering

- Technical Indicators: RSI, MACD, MACD Signal, Bollinger Bands (upper/lower/width), ATR, OBV, EMA (10/20/50/200-day), SMA (10/20/50-day), Stochastic Oscillator (%K, %D), ADX, CCI, Williams %R
- Lag Features: closing price lagged by 1, 2, 3, 5, 10, 20 trading days; lagged volume; lagged returns
- Rolling Statistical Features: 5/10/20-day rolling mean, standard deviation, minimum, maximum, and skewness of closing price and volume
- Calendar & Regime Features: day of week, month, quarter, earnings season flag, VIX-based market regime indicator
- Volume-Price Features: volume-price divergence, on-balance volume momentum, price-to-volume ratio
- Sentiment Features: daily aggregated FinBERT positive, negative, and neutral probability scores; net sentiment score; article count per day
- Interaction Features: sentiment score × 5-day price momentum; sentiment score × daily trading volume; net sentiment × RSI; net sentiment × MACD
- Return-based Features: 1-day, 5-day, and 20-day log returns; realized volatility (rolling 20-day standard deviation of returns)

Financial Validation & Backtesting

- Profit & Loss (P&L) Simulation: simulate trades based on predicted buy/sell signals over the test period to measure gross and net returns

- Sharpe Ratio: compute risk-adjusted return to compare models not just on accuracy but on investment quality
- Backtesting Framework: walk-forward backtesting on the held-out test set, applying model-generated signals to historical price data to evaluate real-world strategy performance
- Benchmark Comparison: compare all model strategies against a buy-and-hold baseline and a random signal baseline

Evaluation Metrics

- Regression: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE)
- Classification: Directional Accuracy (DA), Precision, Recall, F1-Score, ROC-AUC
- Financial: Total Return (%), Sharpe Ratio, Maximum Drawdown, Win Rate
- Baseline comparison: naïve model (prior-day close) for regression; random classifier and buy-and-hold for buy/sell strategy

Tools & Frameworks

- Python 3.10+ - primary language
- yfinance - data acquisition for price and news data
- Pandas, NumPy - data manipulation
- TA-Lib / pandas-ta - technical indicator computation
- Scikit-learn - preprocessing, Random Forest, evaluation utilities
- PyTorch - LSTM+GRU, BiLSTM, CNN+LSTM, and Time Series Transformer implementations
- Hugging Face Transformers - FinBERT sentiment pipeline
- SHAP - feature importance and model explainability for Random Forest
- Matplotlib, Seaborn, Plotly - visualization including backtesting equity curves
- Google Colab / Kaggle Notebooks - GPU-accelerated training environment

3.2 Excluded Components

- Real-time or live trading deployment - the system will not be connected to any brokerage API or execute real trades.
- Production APIs or web services - no REST endpoints or dashboards will be developed for external consumption.
- Options, futures, or derivatives pricing - scope is limited to common equity (stock) prediction only.
- Cryptocurrency markets - different market dynamics are outside this project’s scope.
- Macroeconomic indicators (GDP, CPI, Federal Reserve rate data) - excluded from the feature space; acknowledged as future work.
- Social media sentiment (Reddit, Twitter/X) - only Yahoo Finance News is used; other sentiment sources are future work.
- Portfolio optimization or position sizing - backtesting uses fixed equal-weight allocation; no Kelly Criterion or mean-variance optimization.
- Mobile or web application development - deliverables are Jupyter notebooks and a written report only.
- Intraday or high-frequency trading - all predictions are daily frequency (next-day closing price and signal).
- Distributed or large-scale computing frameworks (Spark, Dask) - training is performed on a single machine with GPU access.

4. Data Sourcing Plan

4.1 Primary Dataset

Attribute	Details
Source	Yahoo Finance via yfinance Python library (v0.2.x)
Price Data	Historical daily OHLCV: Open, High, Low, Close, Adjusted Close, Volume - for all S&P 500 constituent stocks
News Data	Yahoo Finance News headlines per ticker - used for FinBERT sentiment scoring. Accessed via yfinance.Ticker().news
Time Range	January 2015 - December 2024 (approximately 10 years, ~2,500 trading days per ticker)
Universe	All S&P 500 constituent stocks (approximately 500 tickers). Stocks with fewer than 3 years of data will be excluded.
Volume	~500 tickers × 2,500 days = ~1.25 million rows of price data; thousands of news articles per ticker for sentiment
Access Method	yfinance.download() for bulk price data; yfinance.Ticker(symbol).news for news headlines; no API key required
Cost	Free - no subscription or API key required for standard yfinance access

4.2 Backup Datasets

Source	Data Provided	When to Use
Alpha Vantage API	OHLCV + structured news sentiment endpoint	Primary backup if yfinance rate limits become restrictive. Free tier provides 500 API calls/day, sufficient for phased data collection.
Kaggle: S&P 500 Stocks Dataset	Pre-downloaded OHLCV for all S&P 500 stocks	Offline backup if API access fails entirely. Useful for rapid prototyping without network dependency.
Quandl / Nasdaq Data Link	Historical US equity prices and adjusted close data	Secondary price data backup. Free academic tier available; clean, well-maintained historical data.
NewsAPI.org	Financial news headlines from major outlets (Reuters, Bloomberg, CNBC)	Backup sentiment data source if Yahoo Finance News coverage is insufficient for certain tickers. Free developer tier provides 100 requests/day.

5. Risk Assessment & Mitigation Plan

The following risks have been identified as potential threats to the successful and on-time completion of this project. Each risk is accompanied by a description and a specific mitigation strategy.

Risk 1: Data Availability and API Rate Limiting
Description: Bulk downloading data for approximately 500 S&P 500 tickers via yfinance may trigger API rate limits, resulting in incomplete downloads. Additionally, some tickers - particularly mid- and small-cap stocks - may have missing or sparse Yahoo Finance News articles, producing incomplete sentiment features.
Mitigation: Data will be downloaded in batches with randomized delays between requests to avoid rate limiting. Kaggle S&P 500 datasets and Alpha Vantage API will serve as backup sources. For tickers with no news on a given day, sentiment will be set to neutral (0.5) and flagged with a binary 'no-news' indicator to preserve model integrity.

Risk 2: Model Overfitting
Description: With hundreds of engineered features and years of daily training data, deep learning models such as LSTM+GRU, BiLSTM, and CNN+LSTM may memorize training patterns rather than learning generalisable market behaviour, resulting in poor out-of-sample performance.
Mitigation: Time-series-aware walk-forward cross-validation will be applied to all models to prevent data leakage. Regularisation techniques including dropout layers, early stopping, and L2 weight decay will be used for deep learning models. For Random Forest, max depth and minimum sample constraints will be enforced. Training versus validation performance gaps will be monitored throughout.

Risk 3: Temporal Data Leakage
Description: Accidentally incorporating future information into training features is a well-known risk in financial modelling. For example, computing rolling averages or sentiment scores without proper look-back constraints can allow the model to implicitly "see" future data, producing artificially inflated performance metrics that do not reflect real-world predictive power.
Mitigation: A strict chronological train-test split (60% earliest data for training, 20% validation, 20% most recent for testing) will be enforced throughout. All feature engineering transformations - including rolling statistics, lag features, and sentiment scores - will be computed with a minimum one-day lag to ensure no future information is used as a predictor.

Risk 4: High Computational Cost Across Five Models
Description: Training five distinct model architectures - LSTM+GRU, BiLSTM, CNN+LSTM, Time Series Transformer, and Random Forest - across hundreds of S&P 500 tickers is computationally intensive. Free-tier GPU environments such as Google Colab impose session time limits and memory constraints that may restrict full-scale training.
Mitigation: All five models will initially be developed and tuned on a representative subset of 20 - 30 stocks spanning different sectors. Once pipelines are validated, training will be scaled to the full S&P 500. All pre-computed features and FinBERT sentiment scores will be cached to disk to avoid redundant computation across model runs.

Risk 5: Scope Creep
Description: Given the breadth of this project - five models, advanced feature engineering, backtesting, and financial metrics - there is a risk that the team may attempt to add additional components such as macroeconomic indicators, social media sentiment, or portfolio optimisation beyond the defined scope, jeopardising on-time delivery.
Mitigation: A strict feature and model freeze will be enforced after Week 4. Any ideas beyond the defined scope will be logged as future work in the final report but will not be implemented. A shared task board will be used to track deliverables and lock scope boundaries throughout the project.

6. References

1. A. Farhadi, A. Zamanifar, A. Alipour, A. Taheri, and M. Asadolahi, “A hybrid LSTM-GRU model for stock price prediction,” IEEE Access, vol. 13, pp. 117594 - 117618, Jul. 2025, doi: https://doi.org/10.1109/access.2025.3586558.

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5. R. Aroussi, “GitHub - ranaroussi/yfinance: Yahoo! Finance market data downloader (+faster Pandas Datareader),” GitHub - https://github.com/ranaroussi/yfinance